

John Lucaccionni

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Education & Certifications

Indiana University | Luddy School of Informatics, Computing, and Engineering | Bloomington, IN **May 2026**
B.S. Degree, Data Science | Minor: Computer Science **3.689/4.00 GPA**
Honors: Dean's List for the Luddy School, Dean's Scholarship, Luddy Dean's Academic Merit Scholarship
Certificates: AWS Cloud Practitioner
Relevant Coursework: Statistics, Data Analysis & Mining, Database Concepts

Relevant Work Experience

Data Analyst | Indiana University Office of School Partnerships, Bloomington, IN **June 2024-Present**
• Analyzed Indiana high school engagement data by building an interactive, layer-controlled map to identify geographic coverage, participation gaps, and partnership reach for campus leadership.
• Developed visualizations to communicate engagement trends and geographic patterns for campus leadership strategic planning.
• Improved analytical accuracy by standardizing school and district records with unique state education identifiers.
• Designed school and district data schemas and integrated IDOE dataset through ingestion scripts for analysis of engagement trends.
• Established data entry and tagging standards to ensure consistent reporting and long-term analytical usability.

Enterprise Architecture Internship | Tenneco, Northville, MI **June 2025 – August 2025**
• Analyzed SAP audit reporting workflows and automated data extraction and validation, reducing reporting time by 98% (1 week → 30 minutes) and eliminating more than 40 hours of manual effort required for each generated report.
• Improved data accessibility by designing a GenAI-powered knowledge base that enabled plain-English queries, allowing nontechnical teams to retrieve, interpret, and use structured business data independently without requiring SQL expertise.
• Integrated enterprise systems with Azure Logic Apps to improve contract visibility and reduce manual reconciliation.
• Provisioned AWS (EC2, RDS, Redshift) with Terraform and built Power BI dashboards for stakeholder insights.

Relevant Project Experience

Data Science Capstone | Clinical EHR Data Quality Analysis & Reliability Assessment, Bloomington, IN **August 2025-Present**
• Developed a peer-relative data quality assessment framework to evaluate EHR documentation completeness across a statewide wound care registry standardized to the OMOP Common Data Model.
• Built SQL/Python data pipelines to extract, filter, and engineer clinical features from PostgreSQL OMOP database.
• Applied KNN-based peer benchmarking to identify anomalous documentation gaps and prioritized review cohorts for stakeholders.
• Validated model robustness through permutation testing, sensitivity analysis, and concordance with established clinical guidelines.
• Identified systematic racial disparities in documentation completeness after controlling for patient complexity, revealing completeness gaps not detectable through descriptive statistics alone.

Predictive Analysis Project | NHL Game Outcome Analysis, Bloomington, IN **November 2025- December 2025**
• Analyzed NHL game data from a public API by cleaning, structuring, and validating records for exploratory analysis.
• Engineered and examined features such as team performance trends, rest days, and matchup context using a time-aware train/validation/test split to reflect real-world forecasting conditions.
• Compared KNN and Random Forest models (Python, scikit-learn) to evaluate interpretability and predictive performance.
• Evaluated model results to assess generalization and summarize insights and limitations.

Decision Analysis Project | Referee Schedule Builder, Naperville, IL **June 2024- July 2024**
• Analyzed availability, game schedules, and location constraints to develop a decision-support tool that optimizes game selection.
• Aggregated and analyzed multi-source scheduling data from external APIs to enable comparison and informed decision-making.
• Designed a scheduling algorithm that analyzed time conflicts, travel distance, and transition feasibility to optimize game selection.
• Cut scheduling time by 2 hours a week and increased earnings by 20% through automated recommendations.

Software Development | Tranquility App, Bloomington, IN **April 2024- May 2024**
• Built a native iOS application in Swift delivering a customizable ambient-sound experience through a sleek, intuitive user interface.
• Led the end-to-end app development lifecycle, from requirements and UML-based design to implementation and deployment.
• Implemented a modular audio mixer enabling users to blend ambient sounds to create personalized soundscapes.

Technical Skills

- **Data Analysis & Visualizations:** SQL, Python (Pandas, NumPy), R, Tableau, Power BI
- **Data Management & Integration:** PostgreSQL, Data Cleaning, Data Pipelines, API Integration
- **Cloud & Automation:** AWS (EC2, RDS, Redshift, Bedrock), Terraform, Azure Logic Apps
- **Programming:** Python, Java, Swift, HTML, Git, GitHub